



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°1_ – RÉSEAUX 2

COURS: RÉSEAUX 2

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L'objectif de ce TD est de:

1. Comprendre le rôle et le fonctionnement de Wireshark.
2. Capturer et analyser le trafic réseau sur une interface donnée.
3. Identifier différents protocoles et comprendre leurs interactions.
4. Détecter des problèmes simples de réseau et analyser des communications réseau.
5. Comprendre comment Wireshark peut aider à détecter des intrusions réseau.
6. Identifier des comportements suspects ou anormaux sur un réseau.
7. Analyser différents types d'attaques réseau simples : scans de ports, ping flood, tentatives d'intrusion HTTP.

1. Reproduire les taches

1. Ping

```
C:\Users\Lens Sandro Petiote>ping google.com

Pinging google.com [142.251.34.142] with 32 bytes of data:
Reply from 142.251.34.142: bytes=32 time=92ms TTL=111
Reply from 142.251.34.142: bytes=32 time=114ms TTL=111
Reply from 142.251.34.142: bytes=32 time=87ms TTL=111
Reply from 142.251.34.142: bytes=32 time=152ms TTL=111

Ping statistics for 142.251.34.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 87ms, Maximum = 152ms, Average = 111ms

C:\Users\Lens Sandro Petiote>ping google.com

Pinging google.com [142.251.34.142] with 32 bytes of data:
Reply from 142.251.34.142: bytes=32 time=63ms TTL=111
Reply from 142.251.34.142: bytes=32 time=63ms TTL=111
Reply from 142.251.34.142: bytes=32 time=88ms TTL=111
Reply from 142.251.34.142: bytes=32 time=111ms TTL=111

Ping statistics for 142.251.34.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 63ms, Maximum = 111ms, Average = 81ms

C:\Users\Lens Sandro Petiote>
```

icmp									
No.	icmp	Source	Destination	Protocol	Length	Info			
22...	icmpv6	53800 192.168.43.160	142.251.34.142	ICMP	74	Echo (ping) request id=0x0001, seq=131/33536, ttl=128 (reply in 2259)			
22...	168.85859200	142.251.34.142	192.168.43.160	ICMP	74	Echo (ping) reply id=0x0001, seq=131/33536, ttl=111 (request in 2254)			
22...	169.769771200	192.168.43.160	142.251.34.142	ICMP	74	Echo (ping) request id=0x0001, seq=132/33792, ttl=128 (reply in 2264)			
22...	169.884265600	142.251.34.142	192.168.43.160	ICMP	74	Echo (ping) reply id=0x0001, seq=132/33792, ttl=111 (request in 2263)			
22...	170.774011000	192.168.43.160	142.251.34.142	ICMP	74	Echo (ping) request id=0x0001, seq=133/34048, ttl=128 (reply in 2267)			
22...	170.861484900	142.251.34.142	192.168.43.160	ICMP	74	Echo (ping) reply id=0x0001, seq=133/34048, ttl=111 (request in 2265)			
22...	171.778013200	192.168.43.160	142.251.34.142	ICMP	74	Echo (ping) request id=0x0001, seq=134/34304, ttl=128 (reply in 2276)			
22...	171.938020800	142.251.34.142	192.168.43.160	ICMP	74	Echo (ping) reply id=0x0001, seq=134/34304, ttl=111 (request in 2275)			
49...	1239.0058210...	192.168.43.160	142.251.34.142	ICMP	74	Echo (ping) request id=0x0001, seq=135/34560, ttl=128 (reply in 49655)			
49...	1239.0692732...	142.251.34.142	192.168.43.160	ICMP	74	Echo (ping) reply id=0x0001, seq=135/34560, ttl=111 (request in 49654)			
49...	1240.0106747...	192.168.43.160	142.251.34.142	ICMP	74	Echo (ping) request id=0x0001, seq=136/34816, ttl=128 (reply in 49659)			
49...	1240.0738789...	142.251.34.142	192.168.43.160	ICMP	74	Echo (ping) reply id=0x0001, seq=136/34816, ttl=111 (request in 49658)			
49...	1241.0159632...	192.168.43.160	142.251.34.142	ICMP	74	Echo (ping) request id=0x0001, seq=137/35072, ttl=128 (reply in 49666)			
49...	1241.1041557...	142.251.34.142	192.168.43.160	ICMP	74	Echo (ping) reply id=0x0001, seq=137/35072, ttl=111 (request in 49664)			
49...	1242.0197491...	192.168.43.160	142.251.34.142	ICMP	74	Echo (ping) request id=0x0001, seq=138/35328, ttl=128 (reply in 49680)			
49...	1242.1312246...	142.251.34.142	192.168.43.160	ICMP	74	Echo (ping) reply id=0x0001, seq=138/35328, ttl=111 (request in 49675)			

> Frame 2276: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{...}									
> Ethernet II, Src: a2:4f:85:f5:f2:e8 (a2:4f:85:f5:f2:e8), Dst: LiteonTechno_ba:6b:6f (70:08:94:ba:6b:6f)									
> Internet Protocol Version 4, Src: 142.251.34.142, Dst: 192.168.43.160									
> Internet Control Message Protocol									

0000	70	08	94	ba	6b	6f	a2	4f	85	f5	f2	e8	08	00	45	60	p...	ko	0	E'
0010	00	3c	00	00	00	00	6f	01	ad	8f	8e	fb	22	8e	c0	a8	<....o	"		
0020	2b	a0	00	00	54	d5	00	01	00	86	61	62	63	64	65	66	+...T...	..abc	def	
0030	67	68	69	6a	6b	6c	6d	6e	6f	70	71	72	73	74	75	76	ghijklm	opqrstuv		
0040	77	61	62	63	64	65	66	67	68	69							wabc	defg	hi	

2. Téléchargement

tcp									
No.	tcp	Source	Destination	Protocol	Length	Info			
2...	tcp	192.168.43.160	192.168.43.160	TLSv1.3	78	Application Data			
2...	tcp.options.ack_ecn	150.171.109.1...	150.171.109.1...	TCP	54	52429 → 443 [ACK] Seq=8710 Ack=8675 Win=65280 Len=0			
2...	tcp.options.ao	150.171.109.1...	150.171.109.1...	TCP	54	52429 → 443 [FIN, ACK] Seq=8710 Ack=8675 Win=65280 Len=0			
2...	tcp.options.cc	192.168.43.160	192.168.43.160	TCP	54	443 → 52429 [FIN, ACK] Seq=8675 Ack=8710 Win=86016 Len=0			
2...	tcp.options.ccecho	150.171.109.1...	150.171.109.1...	TCP	54	52429 → 443 [ACK] Seq=8711 Ack=8676 Win=65280 Len=0			
2...	tcp.options.cnew	52.85.78.8	192.168.43.160	TCP	55	[TCP Keep-Alive] 61708 → 443 [ACK] Seq=2945 Ack=2042 Win=65024 Len=1			
2...	tcp.options.echo	192.168.43.160	192.168.43.160	TCP	54	[TCP Dup ACK 4601#1] 443 → 52429 [ACK] Seq=8676 Ack=8710 Win=86016 Len=0			
2...	tcp.options.echoreply	150.171.109.1...	150.171.109.1...	TCP	66	[TCP Keep-Alive ACK] 443 → 61708 [ACK] Seq=2042 Ack=2946 Win=74240 Len=0 SLE=2945 SRE=2946			
2...	tcp.options.experimental	52.108.0.37	192.168.43.160	TCP	54	[TCP Retransmission] 52429 → 443 [FIN, ACK] Seq=8710 Ack=8676 Win=65280 Len=0			
2...	tcp.options.md5	52.108.0.37	192.168.43.160	TCP	1454	51763 → 443 [ACK] Seq=230096 Ack=114751 Win=2042 Len=1400 [TCP PDU reassembled in 27205]			
2...	tcp.options.mss	192.168.43.160	192.168.43.160	TCP	1099	Application Data			
2...	tcp.options.nop	54	443	→ 52429 [ACK]	Seq=8676	Ack=8711	Win=86016	Len=0	
2...	tcp.options.qs	192.168.43.160	192.168.43.160	TCP	54	443 → 51763 [ACK] Seq=114751 Ack=8676 Win=2042 Len=0			
2...	tcp.options.rvbd.probe	192.168.43.160	192.168.43.160	TLSv1.2	1404	Application Data			
2...	tcp.options.rvbd.trpy	52.108.0.37	192.168.43.160	TCP	54	51763 → 443 [ACK] Seq=232803 Ack=116101 Win=2042 Len=0			
2...	tcp.options.sack								
2...	tcp.options.sack_perm								
2...	tcp.options.scps								

> Frame 2276: Packet, 74 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{...}									
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0000	a2	4f	85	f5	f2	e8	70	08	94	ba	6b	6f	08	00	45	60	0...			
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3. Application

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 80 || udp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
22..	167.875178500	192.168.43.160	52.110.1.53	TCP	54	62213 → 443 [ACK] Seq=1 Ack=364 Win=251 Len=0
22..	170.851757100	192.168.43.160	135.234.174.40	TCP	55	[TCP Keep-Alive] 52595 → 443 [ACK] Seq=67 Ack=60 Win=254 Len=1
22..	171.008934000	135.234.174.40	192.168.43.160	TCP	66	[TCP Keep-Alive ACK] 443 → 52595 [ACK] Seq=60 Ack=68 Win=299 Len=0 SLE=67 SRE=68
22..	171.504179400	192.168.43.160	52.108.0.37	TCP	1454	51763 → 443 [ACK] Seq=97453 Ack=48601 Win=2047 Len=1400 [TCP PDU reassembled in 2270]
22..	171.504179400	192.168.43.160	52.108.0.37	TLSv1.2	1099	Application Data
22..	171.505044200	192.168.43.160	52.108.0.37	TLSv1.2	316	Application Data
22..	171.726455200	52.108.0.37	192.168.43.160	TCP	54	443 → 51763 [ACK] Seq=48601 Ack=100160 Win=2047 Len=0
22..	171.726455200	52.108.0.37	192.168.43.160	TLSv1.2	1404	Application Data
22..	171.766379000	192.168.43.160	52.108.0.37	TCP	54	51763 → 443 [ACK] Seq=100160 Ack=49951 Win=2042 Len=0
22..	172.137890100	57.144.197.32	192.168.43.160	TLSv1.2	940	Application Data
22..	172.177561300	192.168.43.160	57.144.197.32	TCP	54	52684 → 443 [ACK] Seq=1505 Ack=8074 Win=1020 Len=0
22..	172.221465500	192.168.43.160	57.144.197.32	TLSv1.2	148	Application Data
22..	172.340190500	57.144.197.32	192.168.43.160	TCP	54	443 → 52684 [ACK] Seq=8074 Ack=1599 Win=5407 Len=0
22..	174.268552900	192.168.43.160	192.178.219.1	TCP	55	[TCP Keep-Alive] 52596 → 5228 [ACK] Seq=1 Ack=1 Win=255 Len=1
22..	174.490621100	192.178.219.188	192.168.43.160	TCP	66	[TCP Keep-Alive ACK] 5228 → 52596 [ACK] Seq=1 Ack=2 Win=1047 Len=0 SLE=1 SRE=2
22..	175.368878400	192.168.43.160	52.110.1.91	TLSv1.2	100	Application Data
22..	175.617029000	52.110.1.91	192.168.43.160	TCP	54	443 → 58842 [ACK] Seq=695 Ack=415 Win=254 Len=0

> Frame 2274: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF...
 > Ethernet II, Src: LiteonTechno_ba:6b:6f (70:08:94:ba:6b:6f), Dst: a2:4f:85:f5:f2:e8 (a2:4f:85:f5:f2:e8)
 > Internet Protocol Version 4, Src: 192.168.43.160, Dst: 52.108.0.37
 > Transmission Control Protocol, Src Port: 51763, Dst Port: 443, Seq: 100160, Ack: 49951, Len: 0

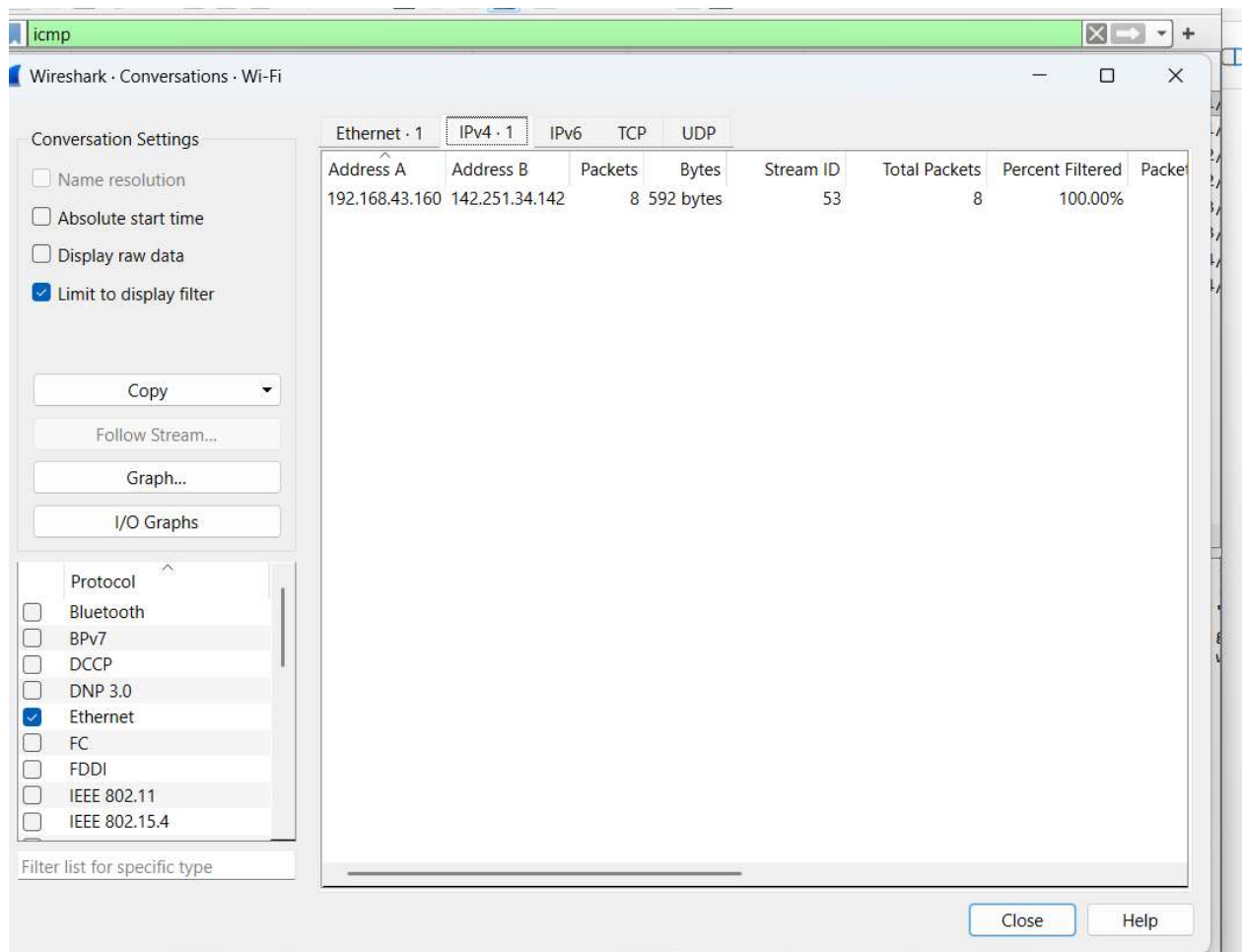
0000 a2 4f 85 f5 f2 e8 70 08 94 ba 6b 6f 08 00 45 00 -0...p...ko..E-
 0010 00 28 e6 de 40 00 80 06 f3 17 c0 a8 2b a0 34 6c -(.@...+..+41
 0020 00 25 ca 33 01 bb 7e bf 65 e4 44 3f 6f f9 50 10 -%3...e-D?oP-
 0030 07 fa 22 36 00 00 --"6..

2. Protocol Hierarchy

Wireshark - Protocol Hierarchy Statistics - Wi-Fi

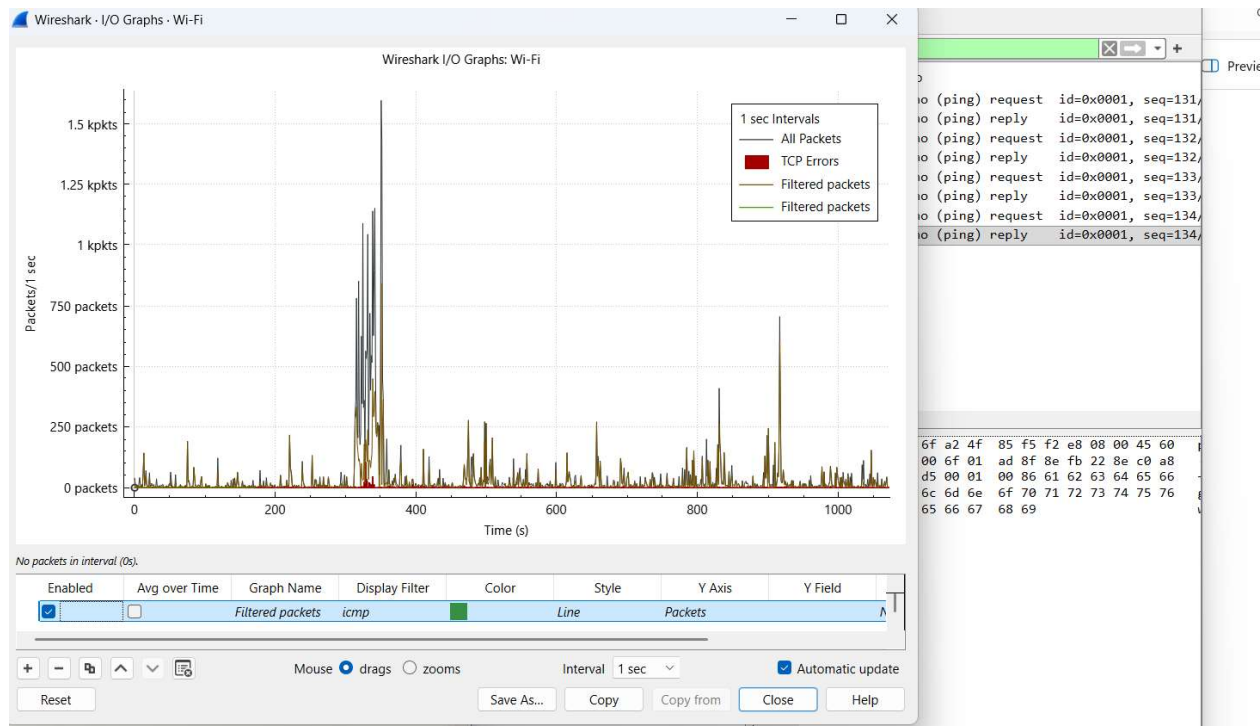
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	16	100.0	1184	8	0	0	0	16
Ethernet	100.0	16	18.9	224	1	0	0	0	16
Internet Protocol Version 4	100.0	16	27.0	320	2	0	0	0	16
Internet Control Message Protocol	100.0	16	54.1	640	4	16	640	4	16

3. Conversations



- L'IP qui envoie le plus de paquets est 192.168.43.160

4. I/O Graphs



Coclusion.

Ce TD m'a permis de comprendre le rôle fonctionnement de Wireshark, analyser le trafic réseau sur une interface donnée, détecter des problèmes simples de réseau et analyser des communications réseau et analyser différents types d'attaques réseau simples. donc la tache a été réussie.